



Universidad
Indoamérica

UNIT 9 · FUNDAMENTALS OF OPERATING SYSTEMS

The model proposes, the operating system disposes

Universidad Tecnológica Indoamérica

AI-native OS

From chat to agent

LLM

Text → next text.

Agent

Text → **action** (tool) → result → text...

- It acts on behalf of the user, but **it is not the user**.
- It can be confidently wrong or be tricked (prompt injection).

Five responsibilities, three platforms

delete old_report.pdf

TD8 (this course)

user

agent (LLM)

tool call

OS: decides

file / app

Agent identity

UNIX socket 0600 + SO_PEERCRED (client uid and pid)

Tools (connectors)

closed catalogue of 11 tools; strict parser

Consent

agentd asks before delete_file

Isolation

fork + exec of agent_exec: setrlimit, alarm and Landlock

Auditing

audit.log with O_APPEND: one line per decision

Windows as an agentic OS

- **Agent workspace** with the agent's own account (preview)
- **Agent connectors**: MCP servers in the **On-Device Registry** (Insiders, Dec. 2025)
- Principles: distinguishable actions, user authorisation, least privilege
- **Microsoft Execution Containers** (Build 2026): process · session · micro-VM

Sources and dates: manual M9 §2.

Linux: pieces to assemble

Landlock

paths and network, without root

seccomp

which calls

namespaces

· cgroups

what it sees · how much it uses

Distributions: Ubuntu announces *inference snaps* and confined agents (previews in 26.10).

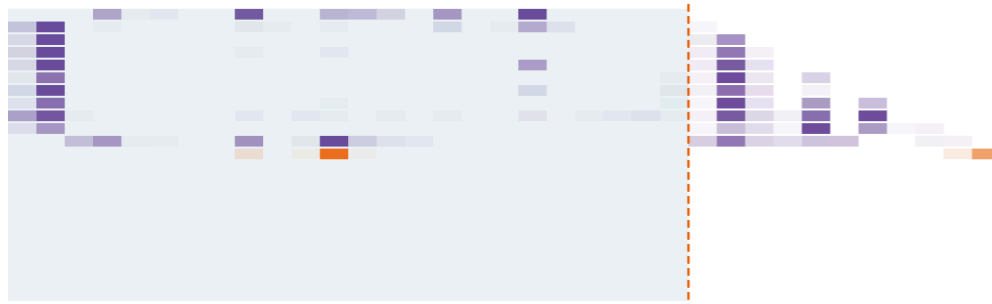
Hardware: the `accelel` subsystem for NPUs.

An LLM inside: bytes, KV cache, attention

request and answer, byte by byte

Q : _ s h o w _ m e _ n o t a s . t x t ↵ A : _ r e a d _ f
i l e ("

attention map: one row per step, one column per byte



next byte: 5 most likely

n	100 %
w	0.0 %
y	0.0 %
p	0.0 %
-	0.0 %

```
read_file(")
```

174,656 parameters in C, measured by the OS

Real output

```
$ ./tinylm -v "lista los archivos de docs"
""
list_files("docs")

model      : 174656 parameters, 3 layers, 4 heads, d=64, context 128 bytes
weights    : 698688 bytes mapped with mmap (MAP_PRIVATE, PROT_READ)
KV cache   : 201984 bytes per request (+ scratch)
tokens     : 33 of prompt (prefill) + 19 generated (decode)
time       : load 0.042 ms, inference 9.08 ms (5730 tokens/s)
page faults: 2 during load, 47 during the first inference (minor; major: 0)
```

- The weights are **mapped** (unit 6): loading takes tenths of a millisecond and 2 page faults; the rest come when they are used (unit 4)
- The KV cache is **per-request** memory; the weights are shared by all

Being right 99 % of the time is not enough

v1

99.4 % exact match on its test... but in the lab it turned `viejo.txt` into `viefe.txt`: it had seen few names like that.

v2

The same 3-layer model (174,656 parameters), but data with more varied names and 12,000 steps instead of 10,000: 99.0 % on 500 new requests, and it copies rare names correctly.

1 % of errors over thousands of actions is dozens of wrong actions: that is why the OS **does not trust** the model's output (§5).

Inference, as the OS sees it

PIECE	COURSE UNIT
Weights with <code>mmap</code> , shared	U6
Page faults when touching the weights	U4
KV cache that grows per token	U4
One thread per request in a pool	U7
NPU / GPU as devices	U1 · U3
The agent talks to <code>agentd</code> over a local socket	U8

Inference as an OS workload

micro-LLM (TD8) · measured

tinyllm code

heap: KV cache + scratch

52 / 128 positions · 78.0 KiB

mmap: weights file (read-only)



171 4 KiB pages · 47 faults on use

stack

memory for 8 agent(s)

shared weights (mmap)



every agent copies the weights



weights KV cache

one request: 33 input tokens (prefill) + 19 generated · 9.08 ms in total



≈ 175 μs per token on one core

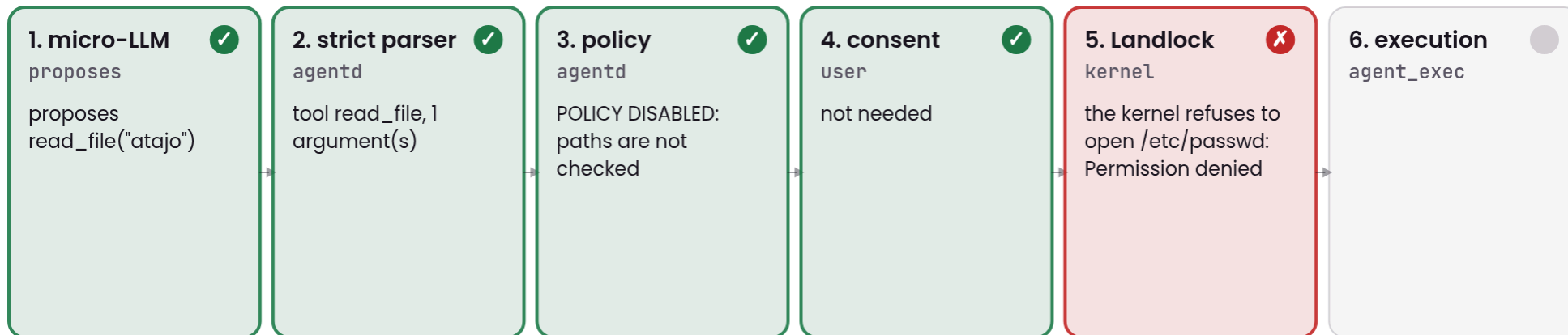
The course's agentic OS

Diagram

```
agent —(UNIX socket 0600)→ agentd — pool of 8 threads
                             | SO_PEERCREC: who is calling (U8)
                             | tinyllm (mmap of weights, KV cache)
                             | strict parser → policy → consent
                             ▼
fork + exec agent_exec
           | setrlimit · alarm · no_new_privs
           | Landlock: only workspace/ and /proc (read-only)
           ▼
           tool → result → audit.log
```

Defence in depth

request: read atajo



line agentd appends to audit.log:

```
2026-09-25T10:00:00 uid=1000 pid=4242 request="read atajo" call=read_file("atajo") decision=deny outcome=error (Permission denied)
```

TD8 + AI challenge

Bash

```
cd tds/bin/td8 && mkdir -p workspace && echo hello > workspace/notes.txt  
./agentd -w workspace &  
./agent "show me notes.txt"  
./agent "delete /etc/passwd"  
cat audit.log
```

Challenge: add `copy_file` (data, retraining, policy, executor) and design 5 attacks.



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TAKE-AWAYS

Three questions

1. Which five responsibilities does an agent demand from the OS?
2. Why is the model output "untrusted text"?
3. Which layer stops the tricky link if the policy fails?